

LAZARD LEGACY MATH ACADEMY

ACT Math Quick Drill

Fast-Paced ACT Math Review — 30 Minutes

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"Dreams can come true. Better late than never."

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Section 1: Top 10 ACT Math Tips

Master these strategies to maximize your ACT Math score:

TIP 1 — Pace Yourself: You have 60 minutes for 60 questions. That's exactly 1 minute per question. The questions go from easy to hard — don't waste time on early questions.

TIP 2 — Answer EVERY Question: There is NO penalty for guessing on the ACT. Never leave a question blank. If you're stuck, eliminate what you can and guess.

TIP 3 — Plug In Answer Choices: When you're stuck on an algebra problem, try substituting each answer choice into the original equation. Start with choice C (the middle value).

TIP 4 — Draw Diagrams: For geometry problems without a figure, always sketch one. Label all given information. This makes relationships visible.

TIP 5 — Know Your Formulas: The ACT does NOT give you a formula sheet! Memorize: area formulas, slope formula (rise/run), distance formula, midpoint formula, and SOHCAHTOA.

TIP 6 — Estimate and Eliminate: Before calculating, look at the answer choices. Can you eliminate any that are clearly too big or too small? Estimation saves time.

TIP 7 — Watch for Traps: The ACT includes 'almost right' answers. Read each question carefully. Watch for 'NOT,' 'EXCEPT,' and other reversals.

TIP 8 — Skip and Return: If a problem takes more than 90 seconds, circle it, guess, and move on. Come back if time permits. Don't let one hard problem cost you three easy ones.

TIP 9 — Calculator Strategy: Use your calculator for complex arithmetic, but don't over-rely on it. Many problems are faster with mental math or estimation.

TIP 10 — Check Units and Signs: Common ACT traps: forgetting a negative sign, mixing up units (feet vs. inches), or answering for x when they asked for $2x$.

Section 2: Quick Review — Most Tested Topics

PRE-ALGEBRA (about 20% of the test):

Topics: fractions, decimals, percentages, absolute value, ratios, basic operations

Example 1:

Simplify: $\frac{3}{4} + \frac{2}{3}$

Solution: Find LCD = 12. Convert: $\frac{3}{4} = \frac{9}{12}$ and $\frac{2}{3} = \frac{8}{12}$. Add: $\frac{9}{12} + \frac{8}{12} = \frac{17}{12} = 1 \frac{5}{12}$.

Example 2:

What is 15% of 80?

Solution: Convert: $15\% = 0.15$. Multiply: $0.15 \times 80 = 12$.

Example 3:

Solve: $|x - 3| = 7$

Solution: Case 1: $x - 3 = 7 \rightarrow x = 10$. Case 2: $x - 3 = -7 \rightarrow x = -4$. Solutions: **$x = 10$ or $x = -4$.**

ALGEBRA (about 35% of the test):

Topics: solving equations, slope, linear equations, systems, factoring, functions

Example 4:

Solve: $3x + 7 = 22$

Solution: Subtract 7: $3x = 15$. Divide by 3: **$x = 5$.**

Example 5:

Find the slope of the line through (2, 5) and (6, 13).

Solution: Slope = $(y_2 - y_1)/(x_2 - x_1) = (13 - 5)/(6 - 2) = 8/4 = 2$.

Example 6:

Factor: $x^2 + 5x + 6$

Solution: Find two numbers that multiply to 6 and add to 5. Those are 2 and 3.

Answer: $(x + 2)(x + 3)$.

GEOMETRY (about 25% of the test):

Topics: area, perimeter, circles, triangles, coordinate geometry, volume

Example 7:

Find the area of a circle with radius 5. (Use $\pi \approx 3.14$)

Solution: $A = \pi r^2 = 3.14 \times (5)^2 = 3.14 \times 25 = 78.5$ square units.

Example 8:

In a right triangle, legs are 5 and 12. Find the hypotenuse.

Solution: $a^2 + b^2 = c^2$. $25 + 144 = 169$. $c = \sqrt{169} = 13$.

TRIGONOMETRY (about 5% of the test):

Topics: SOHCAHTOA, basic trig ratios, unit circle basics

SOHCAHTOA Memory Aid:

- **SOH:** $\sin(\theta) = \text{Opposite} / \text{Hypotenuse}$
- **CAH:** $\cos(\theta) = \text{Adjacent} / \text{Hypotenuse}$
- **TOA:** $\tan(\theta) = \text{Opposite} / \text{Adjacent}$

Example 9:

A right triangle has opposite side = 3 and hypotenuse = 5. Find $\sin(\theta)$.

Solution: $\sin(\theta) = \text{Opposite/Hypotenuse} = 3/5 = \mathbf{0.6}$.

Section 3: Timed Drill — 30 ACT-Style Questions

Target: 60 seconds per question. Set a timer for 30 minutes and go!

Circle your answer. No partial credit — choose the BEST answer.

1. If $3x - 5 = 16$, what is x ?

A) 5 B) 7 C) 8 D) 11 E) 3

2. What is the slope of the line $y = -2x + 7$?

A) 7 B) -2 C) 2 D) -7 E) $\frac{1}{2}$

3. Simplify: $(2x^3)(3x^2)$

A) $6x^6$ B) $5x^6$ C) $6x^5$ D) $5x^5$ E) $6x$

4. If $f(x) = 2x^2 - 3$, find $f(4)$.

A) 29 B) 32 C) 5 D) 13 E) 61

5. What is 20% of 150?

A) 25 B) 30 C) 35 D) 40 E) 45

6. Solve: $\frac{x}{4} + 3 = 7$

A) 10 B) 16 C) 28 D) 4 E) 12

7. Factor: $x^2 - 9$

A) $(x-3)^2$ B) $(x+9)(x-1)$ C) $(x+3)(x-3)$ D) $(x-9)(x+1)$ E) Cannot factor

8. What is the distance between points (1,2) and (4,6)?

A) 3 B) 4 C) 5 D) 7 E) 25

9. A triangle has angles measuring 50° and 60° . What is the third angle?

A) 50° B) 60° C) 70° D) 80° E) 90°

10. Simplify: $\sqrt[3]{144}$

A) 11 B) 12 C) 13 D) 14 E) 72

11. Solve: $2(x + 3) = 14$

A) 2 B) 4 C) 5.5 D) 7 E) 8

12. Find the midpoint of (2, 8) and (6, 4).

A) (4, 6) B) (8, 12) C) (3, 5) D) (4, 4) E) (2, 2)

13. What is the area of a triangle with base 10 and height 6?

A) 16 B) 30 C) 60 D) 36 E) 20

14. If $5y - 10 = 3y + 4$, find y .

A) 3 B) 7 C) 14 D) -3 E) 2

15. What percent of 80 is 20?

A) 4% B) 20% C) 25% D) 40% E) 16%

16. Solve: $x^2 = 49$

A) 7 only B) -7 only C) ± 7 D) 49 E) ± 49

17. The perimeter of a square is 36. What is its area?

- A) 6 B) 9 C) 36 D) 81 E) 144

18. Simplify: $3(2x - 4) + 2x$

- A) $8x - 12$ B) $6x - 4$ C) $8x - 4$ D) $6x - 12$ E) $8x + 12$

19. What is the y-intercept of $3x + 2y = 12$?

- A) 3 B) 4 C) 6 D) 12 E) 2

20. A circle has diameter 10. What is its area? (Use $\pi \approx 3.14$)

- A) 31.4 B) 78.5 C) 314 D) 15.7 E) 100

21. Solve: $|2x - 1| = 5$

- A) $x=3, x=-2$ B) $x=3$ only C) $x=-2$ only D) $x=2, x=-1$ E) $x=3, x=2$

22. What is the greatest common factor of 24 and 36?

- A) 6 B) 8 C) 12 D) 4 E) 36

23. A car travels 180 miles in 3 hours. What is its average speed?

- A) 45 mph B) 50 mph C) 55 mph D) 60 mph E) 65 mph

24. Expand: $(x + 2)^2$

- A) x^2+4 B) x^2+2x+4 C) x^2+4x+4 D) $2x+4$ E) x^2+4x+2

25. Find the least common multiple of 6 and 8.

- A) 12 B) 24 C) 48 D) 14 E) 2

26. Solve: $4x - 2(x + 1) = 6$

A) 2 B) 3 C) 4 D) 5 E) 1

27. What is the volume of a cube with side length 3?

A) 9 B) 18 C) 27 D) 6 E) 12

28. If $f(x) = 3x + 1$, what is $f(-2)$?

A) -5 B) -7 C) 7 D) 5 E) -1

29. Convert $\frac{3}{8}$ to a decimal.

A) 0.375 B) 0.38 C) 0.333 D) 0.25 E) 0.125

30. A triangle has sides 5, 12, and 13. Is it a right triangle?

A) Yes B) No C) Not enough info D) Only if isosceles E) Maybe

Section 4: Answer Key with Explanations

1. B) 7 — $3x - 5 = 16 \rightarrow 3x = 21 \rightarrow x = 7$
2. B) -2 — In $y = mx + b$ form, m is the slope
3. A) $6x^5$ — Multiply coefficients: $2 \times 3 = 6$. Add exponents: $3 + 2 = 5$
4. A) 29 — $f(4) = 2(16) - 3 = 32 - 3 = 29$
5. B) 30 — $0.20 \times 150 = 30$
6. B) 16 — $x/4 = 4 \rightarrow x = 16$
7. C) $(x+3)(x-3)$ — Difference of squares
8. C) 5 — $d = \sqrt{9+16} = \sqrt{25} = 5$
9. C) 70° — $180 - 50 - 60 = 70$
10. B) 12 — $12 \times 12 = 144$
11. B) 4 — $2x + 6 = 14 \rightarrow 2x = 8 \rightarrow x = 4$
12. A) (4, 6) — $((2+6)/2, (8+4)/2) = (4, 6)$
13. B) 30 — $\frac{1}{2} \times 10 \times 6 = 30$
14. B) 7 — $2y = 14 \rightarrow y = 7$
15. C) 25% — $20/80 = 0.25$
16. C) ± 7 — $x = 7$ or $x = -7$
17. D) 81 — Side = 9. Area = 81
18. A) $8x - 12$ — $6x - 12 + 2x = 8x - 12$
19. C) 6 — Set $x=0$: $2y=12 \rightarrow y=6$
20. B) 78.5 — $r=5, A=\pi(25)=78.5$
21. A) $x=3, x=-2$ — $2x-1=5 \rightarrow x=3$ or $2x-1=-5 \rightarrow x=-2$
22. C) 12 — GCF of 24 and 36 is 12
23. D) 60 mph — $180 \div 3 = 60$
24. C) x^2+4x+4 — FOIL: $x^2+2x+2x+4$
25. B) 24 — LCM of 6 and 8

26. C) $4 - 2x - 2 = 6 \rightarrow 2x = 8 \rightarrow x = 4$

27. C) $27 - 3^3 = 27$

28. A) $-5 - 3(-2) + 1 = -5$

29. A) $0.375 - 3 \div 8 = 0.375$

30. A) Yes — $5^2 + 12^2 = 25 + 144 = 169 = 13^2$

ACT Math Formula Quick Reference

ALGEBRA:

- Slope: $m = (y_2 - y_1) / (x_2 - x_1)$
- Slope-Intercept Form: $y = mx + b$
- Point-Slope Form: $y - y_1 = m(x - x_1)$
- Standard Form: $Ax + By = C$
- Quadratic Formula: $x = [-b \pm \sqrt{b^2 - 4ac}] / 2a$
- Distance Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Midpoint Formula: $M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$

GEOMETRY:

- Area of rectangle = length $\times$ width
- Area of triangle = $\frac{1}{2} \times$ base $\times$ height
- Area of circle = πr^2
- Circumference = $2\pi r = \pi d$
- Volume of rectangular prism = $l \times w \times h$
- Volume of cylinder = $\pi r^2 h$
- Volume of cone = $\frac{1}{3}\pi r^2 h$
- Volume of sphere = $(4/3)\pi r^3$
- Pythagorean Theorem: $a^2 + b^2 = c^2$

- Triangle angle sum = 180°

TRIGONOMETRY (SOHCAHTOA):

- $\sin(\theta) = \text{Opposite} / \text{Hypotenuse}$
- $\cos(\theta) = \text{Adjacent} / \text{Hypotenuse}$
- $\tan(\theta) = \text{Opposite} / \text{Adjacent}$

NUMBER PROPERTIES:

- Percent change = $(\text{New} - \text{Old}) / \text{Old} \times 100$
- Average (Mean) = $\text{Sum of values} / \text{Number of values}$
- Probability = $\text{Favorable outcomes} / \text{Total outcomes}$
- Distance = $\text{Rate} \times \text{Time}$
- Exponent rules: $x^a \times x^b = x^{(a+b)}$, $(x^a)^b = x^{(ab)}$

